

Umar Turdumambetov

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github.com/callmeUmar | Portfolio

Professional Summary

Software Engineer and CS student (GPA: 4.0) with hands-on experience developing, deploying, and debugging real-world systems in C++, Python, and Java. Skilled in performance optimization, unit testing, and technical documentation across full-stack applications and backend services. Collaborative, motivated, and comfortable operating in fast-paced environments with changing requirements.

Technical Skills

Languages: C++, C, Python, Java, JavaScript/TypeScript, SQL (PostgreSQL, MySQL), Bash/Shell

Systems & Networking: REST APIs, TCP/IP, Linux, multi-threading, async pipelines, microservices

Testing & Optimization: Unit testing, Postman, performance profiling, debugging, code reviews

Backend & Cloud: FastAPI, Flask, Node.js, AWS (EC2, S3, Lambda), Docker, Kubernetes, GitHub Actions CI/CD

Experience

Researcher

Sep. 2025 – Present

Algoverse (RL & Post-Training Research)

Remote

- Debug and optimize model training pipelines in a fast-paced environment, improving stability and performance by 15–25%.
- Develop evaluation frameworks, unit tests, and documentation, reducing output variance by 20% across diverse tasks.
- Built experimental pipelines to benchmark model improvements and validate robustness across training iterations.

Software Engineer

Nov. 2025 – Dec. 2025

Canva

Remote

- Designed and deployed modular content-generation pipelines in production, boosting throughput by 45% and reducing inconsistencies by 30%.

- Documented system architecture and collaborated cross-team to deliver schemas, improving efficiency by 35%.

Software Engineer

Jun. 2023 – Mar. 2024

International University of Kyrgyzstan

Bishkek, Kyrgyzstan

- Designed and deployed a full-stack app for 200+ users (JavaScript, FastAPI, SQL) integrating RFID and automating 95% of workflows.
- Implemented role-based auth, optimized SQL queries, and enumerated software requirements through documentation and system diagrams.
- Improved system reliability through structured debugging and performance profiling of backend services.

Projects

Churn Predictor — Production System | Deployed on Render

- Designed, developed, and deployed a full-stack system serving **7,043 users** with REST APIs and unit tests — reducing misclassification by **40%**.
- Shipped end-to-end DevOps pipeline (GitHub Actions CI/CD, Docker, automated testing) delivering releases with zero manual intervention.
- Built modular data validation and feature pipelines enabling reproducible experiments and consistent model performance.

PolyPredictor-Kit — 1st Place (QuackHacks 2025) | Devpost

- Engineered a real-time data ingestion system with async FastAPI services achieving sub-**200ms** latency across 50+ concurrent sources.
- Documented architecture and structured **3,000+** records into production datasets, improving signal accuracy by **30%**.
- Optimized concurrency and request handling to support high-frequency prediction workloads with stable throughput.

PromptLock — LLM Context Compression Tool | Devpost

- Designed a two-layer compression system enforcing strict **2K–16K** token budgets with rigorous unit testing across edge cases.
- Reduced failure rates by **65%** and cut inference costs up to **70%** through systematic debugging and performance optimization.
- Implemented adaptive prioritization logic improving context efficiency under constrained inference environments.

AI Visual Novel Creator — 1st Place (CodeDay 2025) | Showcase

- Built a full-stack system with backend APIs and dynamic routing across 20+ states under a 24-hour deadline.
- Cut processing time by **55%** and eliminated **90%+** rendering artifacts through caching and memory profiling.
- Designed scalable state management system enabling dynamic branching and real-time content generation.

Education

Whitman College

May. 2026 – Jun. 2028

Bachelor of Science in Computer Science | **GPA: 4.0**

Walla Walla, WA